

lec520: Final Review I

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1 Ex 1. Predicting House Prices

What is wrong:

- Not standardized - can't compare coefficients by magnitude
- Small sample size - results not generalizable to the population
- Multicollinearity - need to consider interaction terms in regression model
- Outliers/influential points - Skews coefficients in the model, can graph several points for outlier
- Model may not be linear - graph several points to check for linearity

2 Ex 2. Fraud Detection Classification

What is wrong:

- 25% of rows are dropped due to inner merge, ignores potential nonresponse bias - representativeness issue
 - Can instead do an outer merge, replace NaN values with default val, or do analysis on dropped column
- Class imbalance - cases of fraud are much less rare than non-fraud, need to check accuracy, precision, recall of **minority class**
 - Standard `train_test_split` may not provide enough fraud samples in each fold

3 Final Notes

- Final will have more short answer rather than MC